Variable Selection Activity

2026-01-29

SAT example

Key objective: *After accounting for percentage of students who took the test and the median class rank of test-takers (to adjust, somewhat for selection bias in the samples from each state), which variables are associated with state SAT scores?* Note this is slightly different than the example your book worked through, in what way does it differ? Outline what you think would be a reasonable strategy for investigating other variables.

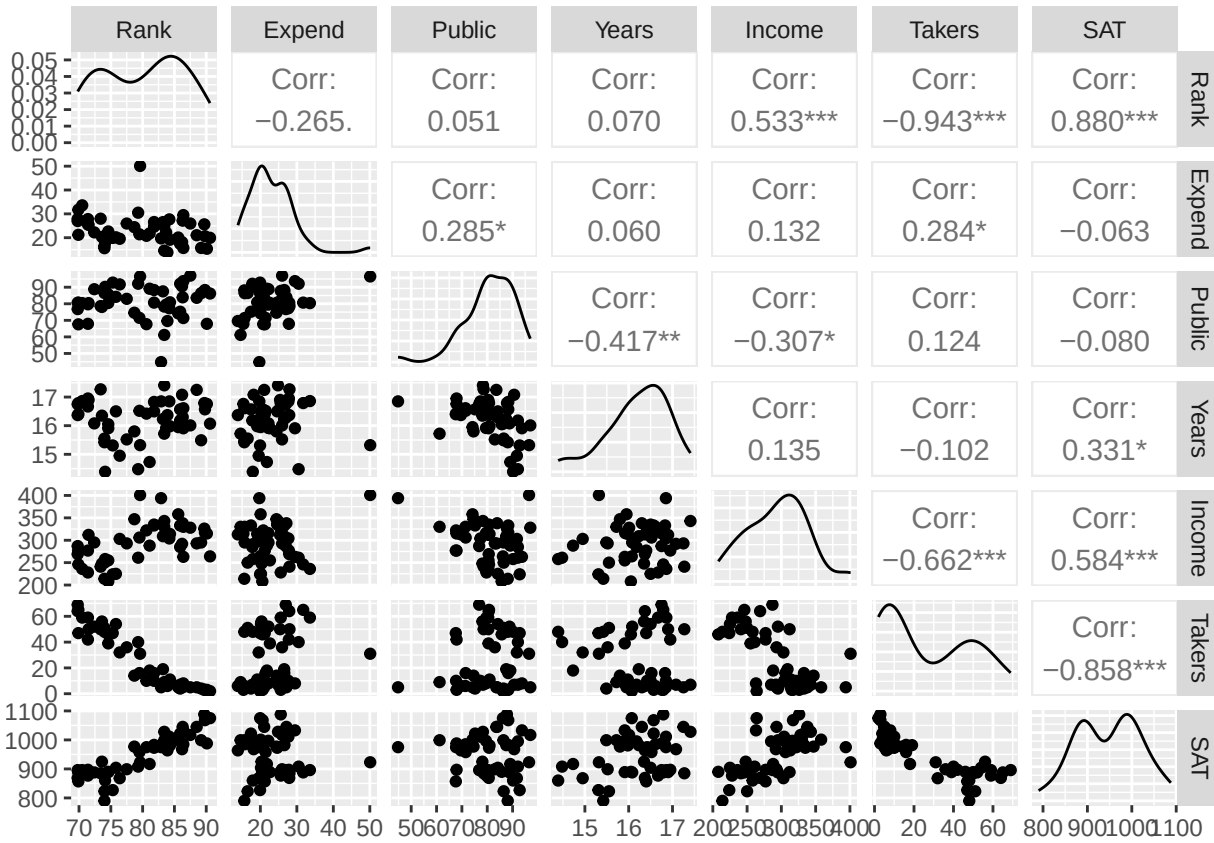
1. Which variables do we want to consider in the variable selection process?

```r
options(show.signif.stars = FALSE)
library(tidyverse)
library(Sleuth3)
sat <- case1201
# ? case1201 and see text for more descriptions of variables
glimpse(sat)
```

```
## Rows: 50
## Columns: 8
## $ State  <fct> Iowa, SouthDakota, NorthDakota, Kansas, Nebraska, Montana, Minn~
## $ SAT    <int> 1088, 1075, 1068, 1045, 1045, 1033, 1028, 1022, 1017, 1011, 100~
## $ Takers <int> 3, 2, 3, 5, 5, 8, 7, 4, 5, 10, 5, 4, 9, 8, 7, 3, 6, 16, 19, 11,~
## $ Income <int> 326, 264, 317, 338, 293, 263, 343, 333, 328, 304, 358, 295, 330~
## $ Years  <dbl> 16.79, 16.07, 16.57, 16.30, 17.25, 15.91, 17.41, 16.57, 16.01, ~
## $ Public <dbl> 87.8, 86.2, 88.3, 83.9, 83.6, 93.7, 78.3, 75.2, 97.0, 77.3, 74.~
## $ Expend <dbl> 25.60, 19.95, 20.62, 27.14, 21.05, 29.48, 24.84, 17.42, 25.96, ~
## $ Rank   <dbl> 89.7, 90.6, 89.8, 86.3, 88.5, 86.4, 83.4, 85.9, 87.5, 84.2, 85.~
```

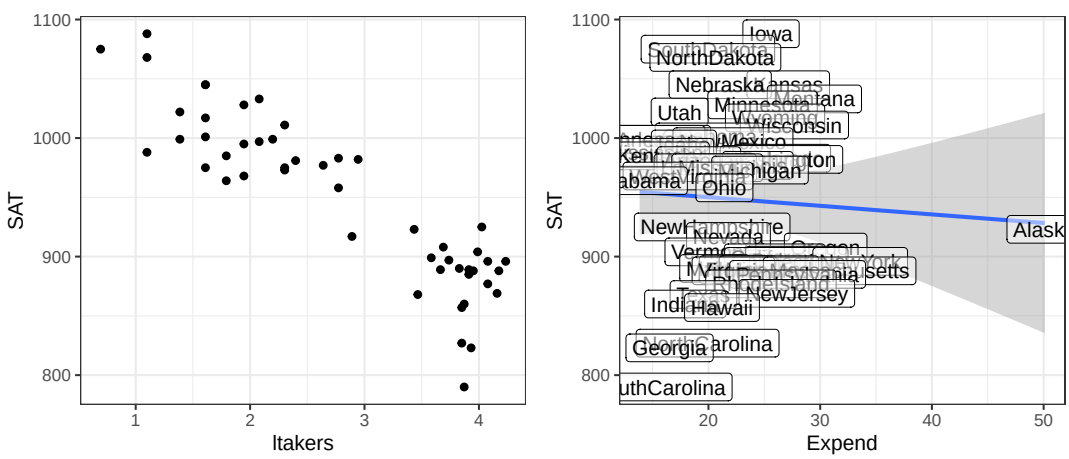
2. Exploratory analysis. What do you see? Would you consider any transformations? Do you notice any potentially influential cases?

```
library(GGally)
library(Sleuth3)
sat <- case1201
ggpairs(sat[,c(8:2)])
```



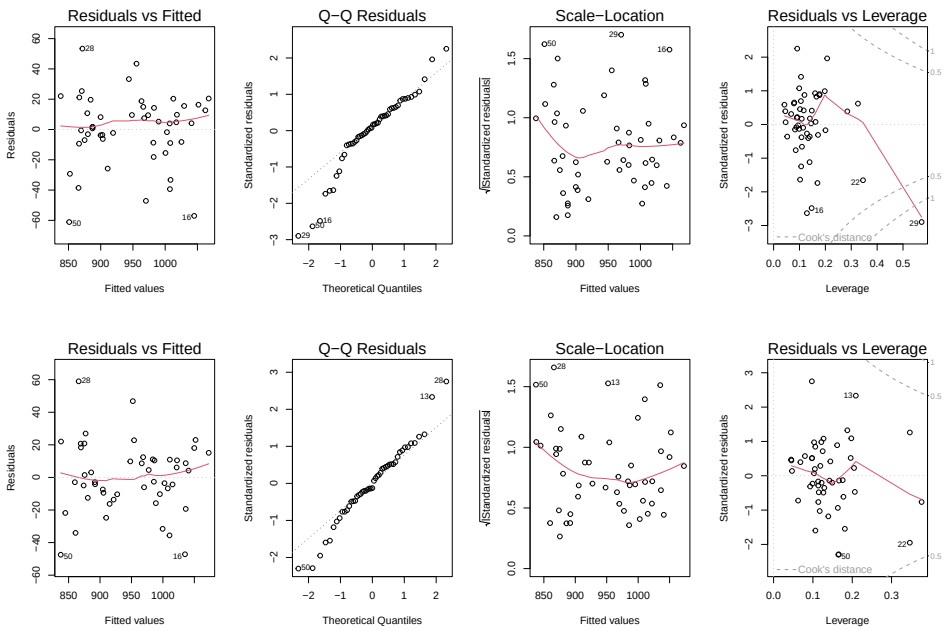
3. Transformations and potentially influential cases suggested by EDA:

```
# note you may need to install patchwork
# install.packages("patchwork")
library(patchwork)
df_sat <- sat |>
  mutate(ltakers = log(Takers))

# create scatterplot for SAT vs. ltakers
p1 <- ggplot(df_sat, aes(x = ltakers, y = SAT)) +
  geom_point() +
  theme_bw()

# create scatterplot for expenditures with labels as state names
p2 <- ggplot(df_sat,
             aes(x = Expend, y = SAT, label = State)) +
  geom_smooth(method = "lm") +
  geom_label(alpha = 0.5)  +
  theme_bw()

# does the relationship between log(takers) and SAT appear more linear?
# does Alaska have high leverage? large residual?
# Do you think it will have high Cook's d?
p1 + p2
```



4. Fit and examine 2 preliminary (rich) models (including AK and excluding AK). If you choose to remove AK from the analysis, how does this impact inferences you can make? Describe in your own words.

```
# compare residual plots with AK
sat_all <- lm(SAT ~ log(Takers) + Rank + Income +
            Years + Public + Expend ,
          data = sat)
# without AK
sat_red <- lm(SAT ~ log(Takers) + Rank + Income +
            Years + Public + Expend,
          data = sat[-29, ])
# plot all in one window
par(mfrow = c(2,4))
plot(sat_all)
plot(sat_red)
```



5. Use sequential VS (forward) to identify variables to include after we've accounted for `log(Takers)` and `Rank` (that is, we're going to force log(Takers) and Rank into all the models we consider). At each step, write down your new "M1" and the new candidate set of models. Repeat until you have a final model (i.e. until the largest $F - to - enter$ is < 4).

```r
# forward selection approach, want to consider all but
# two vars we want in model
m1 <- lm(SAT ~ log(Takers) + Rank,
           data = sat[-29,])
# Mcand1
mcand1 <- lm(SAT ~ log(Takers) + Rank + Expend,
           data = sat[-29,])
anova(m1, mcand1)
```

```
## Analysis of Variance Table
##
## Model 1: SAT ~ log(Takers) + Rank
## Model 2: SAT ~ log(Takers) + Rank + Expend
##   Res.Df   RSS Df Sum of Sq      F    Pr(>F)
## 1     46 45498
## 2     45 24792  1     20706 37.583 1.998e-07
```

```r
# Mcand2
mcand2 <- lm(SAT ~ log(Takers) + Rank + Income,
           data = sat[-29,])
anova(m1, mcand2)
```

```
## Analysis of Variance Table
##
## Model 1: SAT ~ log(Takers) + Rank
## Model 2: SAT ~ log(Takers) + Rank + Income
##   Res.Df   RSS Df Sum of Sq      F Pr(>F)
## 1     46 45498
## 2     45 43899  1    1598.8 1.6389  0.207
```

```r
# ... keep exploring candidate models which variable would you add next? Write
# code to continue the process for all potential Xs
```

This gets rather tedious, so now that you've done it once, you can use the `step` function to complete a similar process. **CAVEAT: `step( )` does not respect hierarchy of interactions or polynomial terms. You may have to adjust some steps.**

6. Consider a forward fit starting with `log(Takers)` and `Rank`. Run the following code. (Note: the direction of selection is specified by the `direction` argument). What criterion is `step` using to make its decisions? When does it decide to stop? How can you calculate the F-to-enter using the available output (note $n = 49$)

```
fit_forward <- step(lm(SAT ~ log(Takers) + Rank,
                       data = sat[-29,]),
                    scope = SAT ~ log(Takers) +
                      Income + Years + Public +
                      Expend + Rank,
                    direction = "forward")
summary(fit_forward)
```

7. Backward selection and stepwise selection can also be quickly implemented with the `AIC`-based inclusion/exclusion criterion using the `step` function. Explore backward and stepwise selection with a first-order SAT. How do your final models compare from backward and stepwise selection if your stopping criterion is that to include/exclude a variable, $\Delta AIC = |AIC_{M1} - AIC_{cand}| > 2$. Explore!

```
fit_backward <- step(lm(SAT ~ log(Takers) + Income + Years +
                                  Public + Expend + Rank,
                        data = sat), direction = "backward")

summary(fit_backward)
```

```
fit_step <- step(lm(SAT ~ log(Takers) + Rank,
                    data = sat),
                 scope = SAT ~ (log(Takers) +
                                  Income + Years +
                                  Public + Expend + Rank),
                 direction = "both")
summary(fit_step)
```

All Subsets regression

The Cp statistic assumes that the "full" model has no bias. The full model considered in the text is:

```
# make dataset that removes alaska
sat2 <- df_sat[-29, ]
full_model <- lm(SAT ~ ltakers + Income + Years +
                   Public + Expend + Rank,
                 data = sat2)
summary(full_model)
```

```
##
## Call:
## lm(formula = SAT ~ ltakers + Income + Years + Public + Expend +
##     Rank, data = sat2)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -47.447 -10.362  -2.626  11.102  59.001
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 287.5240   259.4170   1.108   0.2740
## ltakers     -30.2149    14.7079  -2.054   0.0462
```

```
## Income        0.1029     0.1259   0.817   0.4183
## Years        13.1074     5.8798   2.229   0.0312
## Public       -0.1011     0.5105  -0.198   0.8439
## Expend        3.9367     0.8486   4.639  3.4e-05
## Rank          5.2738     2.2997   2.293   0.0269
##
## Residual standard error: 22.57 on 42 degrees of freedom
## Multiple R-squared:  0.9128, Adjusted R-squared:  0.9003
## F-statistic: 73.28 on 6 and 42 DF,  p-value: < 2.2e-16
```

The `MuMIn` package (Barton, 2017) provides automated model selection procedures (all possible models compared) and a function `AICc` if you want that measure that can be useful for exploratory questions. There are $2^{\#\text{ of }X\text{s}}$ potential models in the all subsets model set. In this case that is $2^6 = 64$ models!

```r
# load MuMIn
library(MuMIn)
# ensure all models use the same data
# need to run to use dredge function
options(na.action = "na.fail")

# defaults to AICc for criterion, set to BIC
allSubsets_sat <- dredge(full_model, rank = "BIC")
```

```
## Fixed term is "(Intercept)"
```

```r
# number of rows is = # models in model set
dim(allSubsets_sat)
```

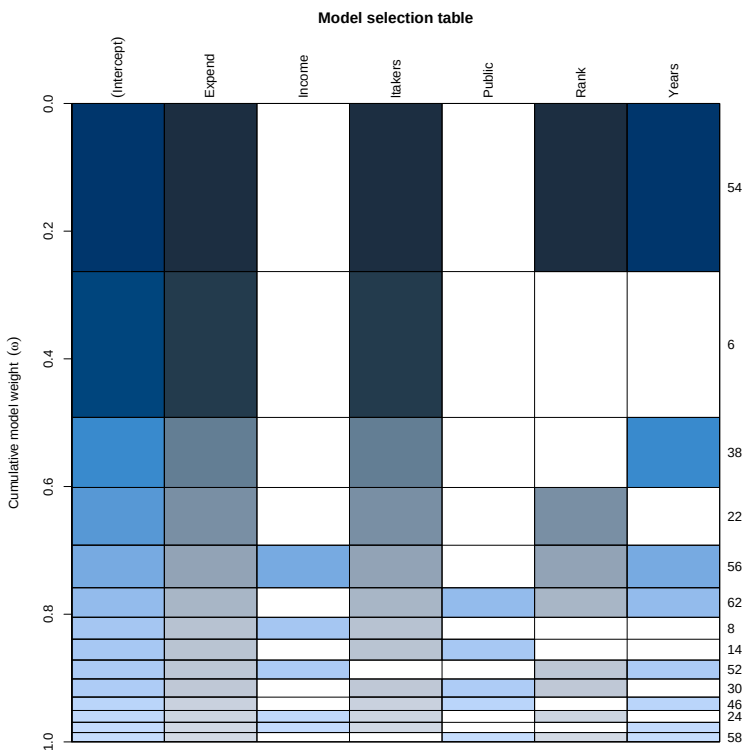
```
## [1] 64 12
```

```r
# look at top set of models with diff in BIC < 6 from "top" model
subset(allSubsets_sat, delta <6)
```

```
## Global model call: lm(formula = SAT ~ ltakers + Income + Years + Public + Expend +
##     Rank, data = sat2)
## ---
## Model selection table
##    (Intrc) Expnd   Incom  ltkrs   Publc   Rank  Years df   logLik   BIC delta
## 54   399.1 3.996         -38.10          4.400 13.150  6 -219.062 461.5  0.00
## 6   1029.0 4.605         -66.17                        4 -223.096 461.8  0.28
## 38   904.9 4.217         -64.63                7.892  5 -221.883 463.2  1.75
## 22   786.7 4.626         -51.60          2.514         5 -222.076 463.6  2.14
## 56   291.2 3.872 0.11350 -31.16          5.060 13.490  7 -218.491 464.2  2.75
## 62   361.3 4.143         -33.73 -0.2785  5.158 12.180  7 -218.854 465.0  3.48
## 8   1017.0 4.570 0.03472 -65.24                        5 -223.045 465.5  4.07
## 14  1030.0 4.611         -66.16 -0.0173                5 -223.095 465.6  4.17
## 52  -255.3 3.385 0.24120                 9.376 18.940  6 -221.248 465.8  4.37
## 30   655.3 4.825         -40.97 -0.5509  4.288         6 -221.288 465.9  4.45
## 46   853.4 4.024         -64.43  0.2906         9.850  6 -221.595 466.5  5.07
## 24   703.8 4.535 0.09619 -46.01          3.031         6 -221.713 466.8  5.30
## 40   899.0 4.200 0.02090 -64.08                7.811  6 -221.864 467.1  5.60
## 58  -166.7 4.076                -0.8503 10.500 15.550  6 -221.951 467.3  5.78
##    weight
## 54  0.263
## 6   0.228
## 38  0.110
## 22  0.090
```

```
## 56  0.067
## 62  0.046
## 8   0.034
## 14  0.033
## 52  0.030
## 30  0.028
## 46  0.021
## 24  0.019
## 40  0.016
## 58  0.015
## Models ranked by BIC(x)
```

```r
# visualzise same summary as above
par(mar = c(3,5,6,4))
plot(subset(allSubsets_sat,delta < 6),
     labAsExpr = TRUE)
```



8. Discuss the output with your group, which model was the "best" with respect to BIC? Does it address the objectives of the study? Do all of the models identified by the all subsets approach include the variables needed to address the question of interest? Answer each of these questions using pieces of the output above.